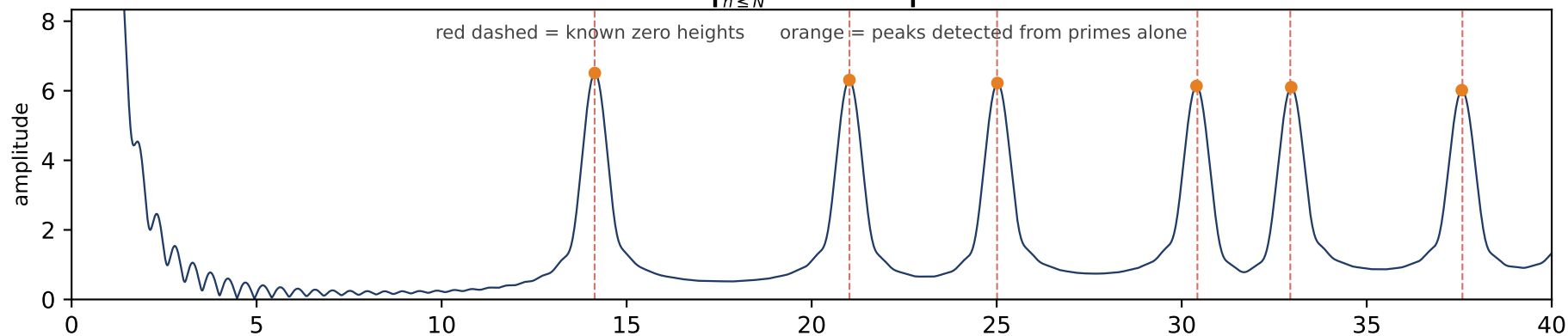
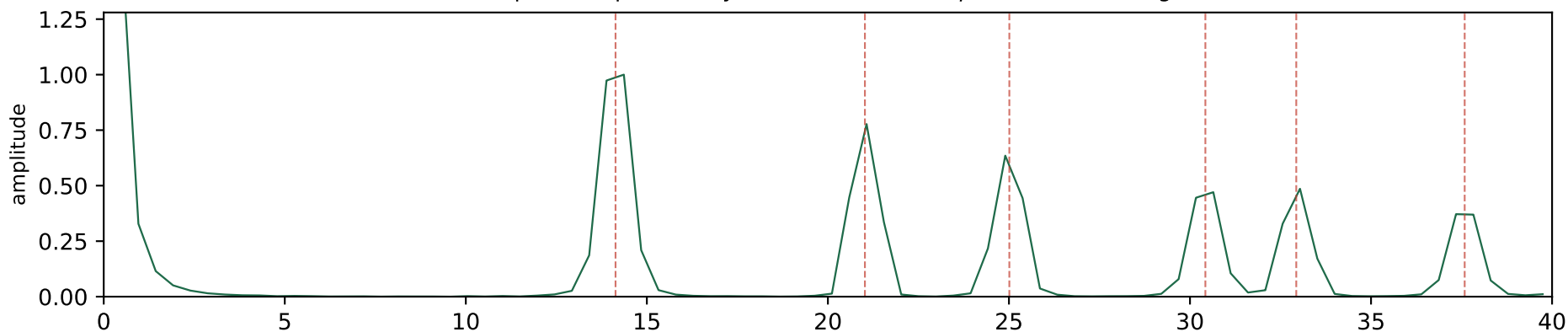


Scale-spectrum of the primes $\left| \sum_{n \leq N} \Lambda(n) n^{-1/2 - i\gamma} \right|$, $N = 10^6$ (ζ never evaluated)



Independent prime-only route: FFT of $e^{-u/2}(\psi(e^u) - e^u)$, $u = \log x$



Resolution = scale-depth: peak at $\gamma_1 = 14.1347$ sharpens as shell N grows (width $\sim 2\pi/\log N$)

